

Organisms and Populations

NEET-UG Botany | Complete NCERT Coverage | Chapter 11, Class 12



Botany for
NEET-UG

Chapter Overview

NCERT Syllabus Mapping

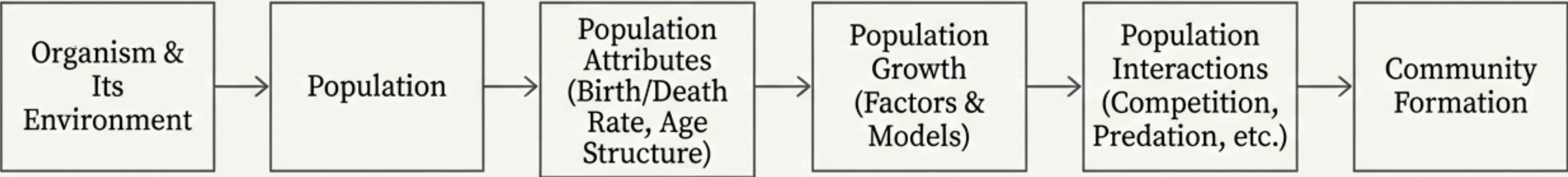
- 11.1 Populations
 - 11.1.1 Population Attributes
 - 11.1.2 Population Growth
 - 11.1.3 Life History Variation
 - 11.1.4 Population Interactions

NEET Weightage Trend



Weightage: HIGH
(Typically 3-4 questions per year, high conceptual importance)

High-Level Concept Flow



11.1.1 Population Attributes (Part 1)

Key Definition: Population

A group of individuals of a given species that live in a well-defined geographical area, share or compete for similar resources, and potentially interbreed.

Examples:

Cormorants in a wetland, teakwood trees in a forest, bacteria in a culture plate.

Core Concept: Attributes of a Population vs. an Individual

Attribute of an Individual	Corresponding Attribute of a Population
Birth	Birth Rate (per capita births)
Death	Death Rate (per capita deaths)
Sex (Male/Female)	Sex Ratio (e.g., 60% female)

Birth Rate

If in a pond there were 20 lotus plants last year and through reproduction 8 new plants are added... we calculate the birth rate as $8/20 = 0.4$ offspring per lotus per year.

Death Rate

If 4 individuals in a laboratory population of 40 fruitflies died during a week... the death rate... is $4/40 = 0.1$ individuals per fruitfly per week.

11.1.1 Population Attributes (Part 2)

Attribute: Age Distribution

A population is composed of individuals of different ages. The graphical representation of the age distribution (% individuals of a given age group) is the **Age Pyramid**.

Shapes of Age Pyramids Reflect Growth Status



- Triangular/Pyramidal → **Expanding (Growing)**



- Bell-shaped → **Stable**



- Urn-shaped → **Declining**

Attribute: Population Size / Population Density (N)

The number of individuals of a species per unit area or volume.

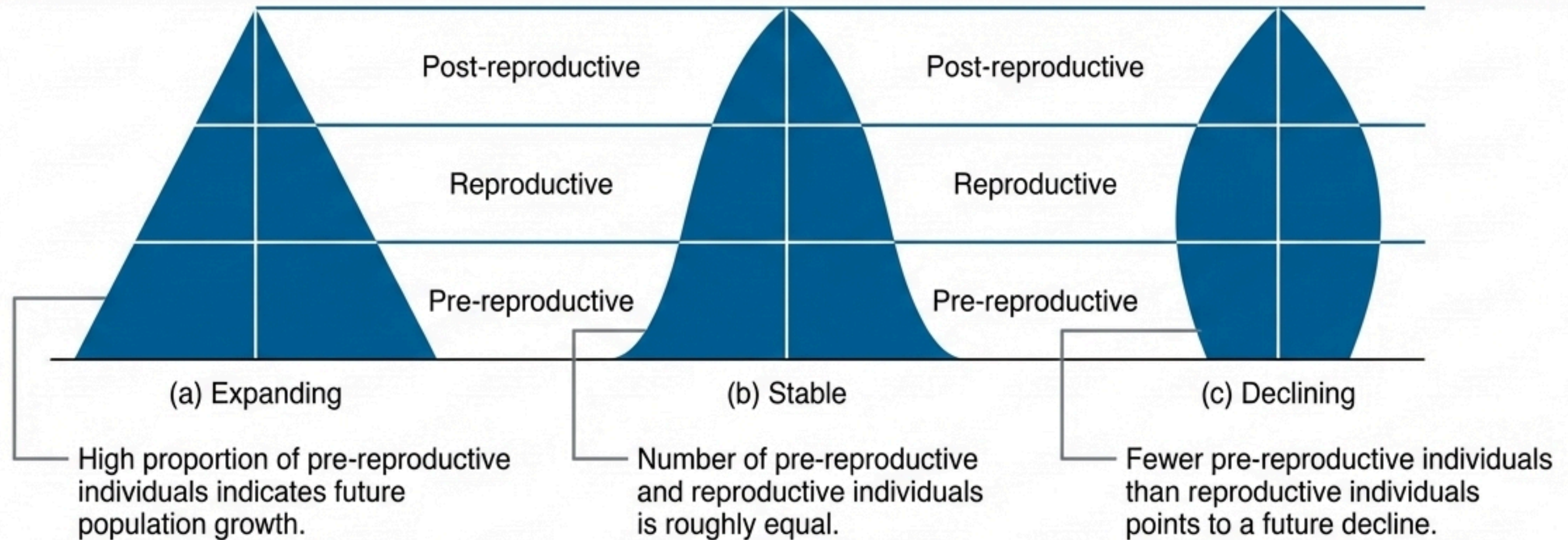
Why absolute numbers aren't always best:

- Meaningless Case: 200 Parthenium plants vs. 1 huge banyan tree. Here, **percent cover** or **biomass** is a more meaningful measure.
- Difficult to Determine: For huge populations (e.g., Chlamydomonas in a pond) or difficult-to-count organisms.

Alternative Measures

- Relative Density: Number of fish caught per trap.
- Indirect Count: Tiger census based on pug marks and fecal pellets.

Visual Explanation: Age Pyramids for Human Population

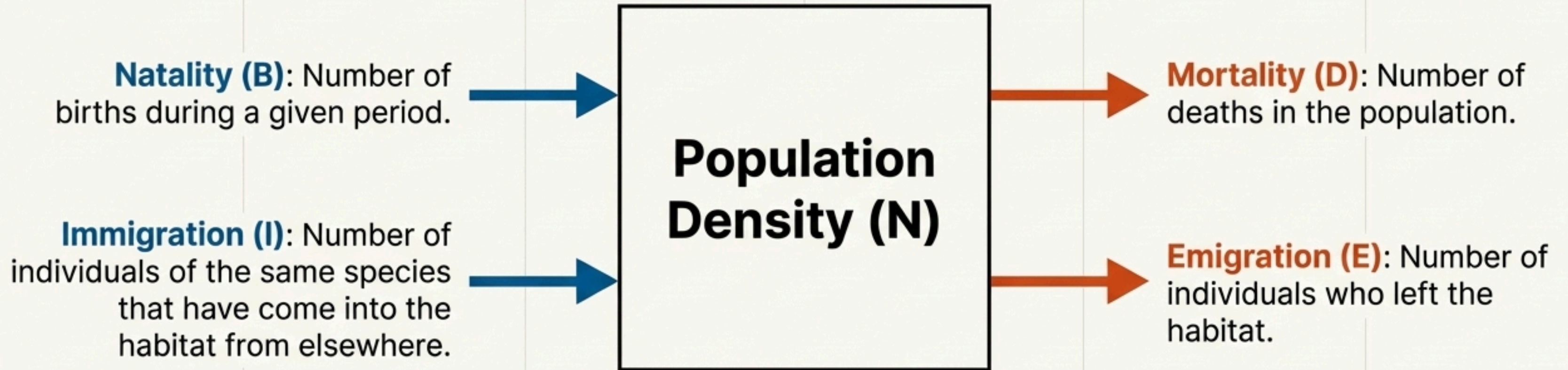


Examiner's Focus

Examiner often asks to identify the growth status based on the shape of the pyramid.

11.1.2 Population Growth: The Four Basic Processes

Population size is not static. Its density fluctuates due to four processes.



$$N_{t+1} = N_t + [(B + I) - (D + E)]$$

Explanation: Population density will increase if $(B + I) > (D + E)$.

NCERT Insight

Under normal conditions, births and deaths are the most important factors. Immigration and emigration assume importance only under special conditions (e.g., colonization of a new habitat).

Growth Models: (i) Exponential Growth

Governing Condition

Unlimited Resources (food and space).
The population realizes its full innate potential to grow.

Key Parameter: Intrinsic Rate of Natural Increase (r)

$r = (b - d)$ where 'b' is per capita birth rate and 'd' is per capita death rate. 'r' is a very important parameter for assessing the impact of any biotic or abiotic factor on population growth.

Equations

Differential Form:

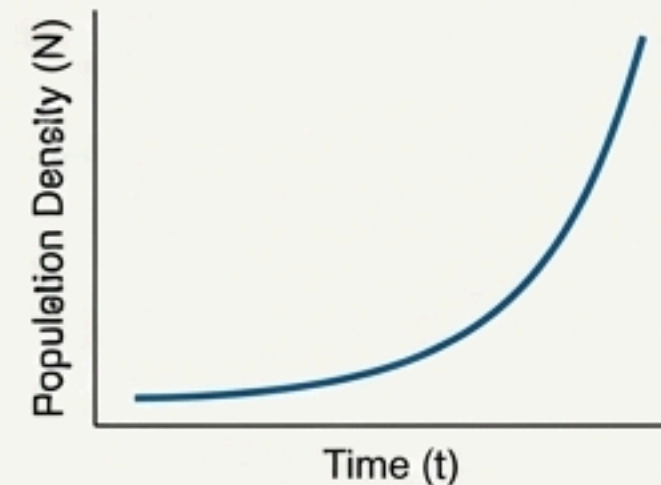
$$\frac{dN}{dt} = rN$$

Integral Form:

$$N_t = N_0 e^{rt}$$

Where: N_t = Population density after time t , N_0 = Density at time zero, r = intrinsic rate, e = base of natural logarithms (2.71828).

Graphical Representation



When we plot N in relation to time, the result is a J-shaped curve.

Growth Models: (ii) Logistic Growth

Limited Resources. This leads to competition between individuals.

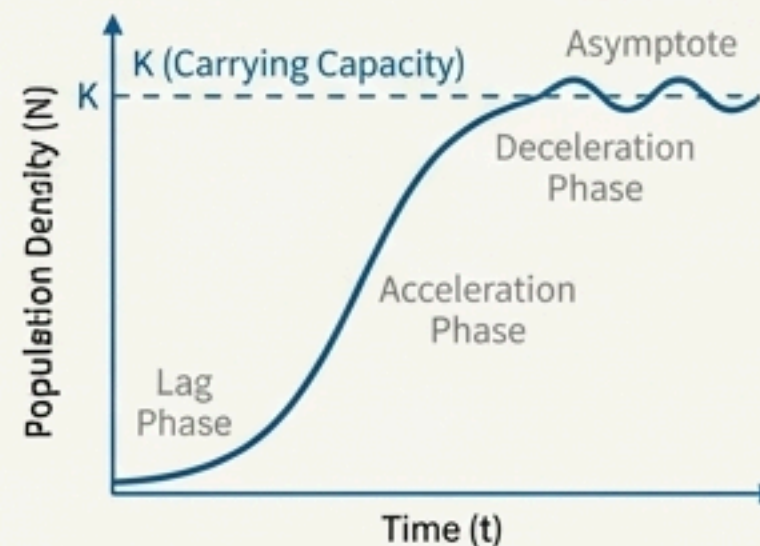
In nature, a given habitat has enough resources to support a maximum possible number, beyond which no further growth is possible. This limit is nature's **carrying capacity (K)**.

Phases of Growth

1. **Lag Phase** (initial slow growth)
2. **Acceleration Phase** (log phase)
3. **Deceleration Phase** (growth slows)
4. **Asymptote** (population density reaches K)

Graphical Representation

A plot of N in relation to time (t) results in a **Sigmoid curve (S-shaped curve)**.

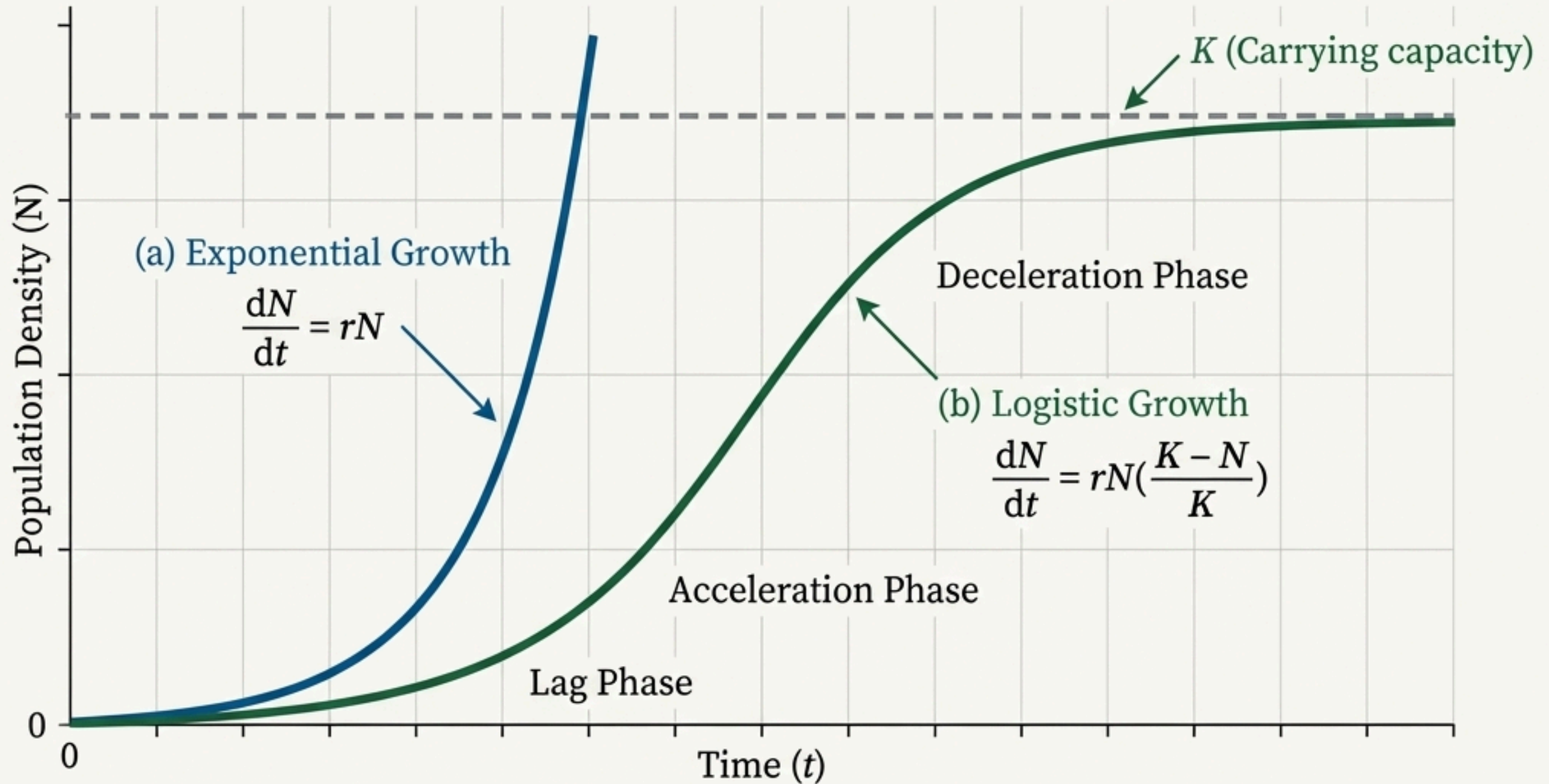


Equation (Verhulst-Pearl Logistic Growth)

$$\frac{dN}{dt} = rN \left(\frac{K - N}{K} \right)$$

Since resources for growth for most animal populations are finite, the logistic growth model is considered a more realistic one.

Visual Explanation: Population Growth Curves



Examiner's Focus

Questions often test the understanding of the equations, the shape of the curves, and the concept of 'K' as a limiting factor.

11.1.4 Population Interactions: An Overview

Core Idea: In nature, animals, plants and microbes do not and cannot live in isolation but interact in various ways to form a biological community.

Interspecific Interactions:

Interactions of populations of two different species.

Classification of Interactions:

Classification of Interactions:

“+” for beneficial interaction, “-” for detrimental interaction, “0” for neutral interaction.

Name of Interaction	Species A	Species B
Mutualism Interspecific Interactions	+	+
Competition	-	-
Predation	+	-
Parasitism	+	-
Commensalism	+	0
Amensalism	-	0

Population Interactions: Predation (+, -) and Competition (-, -)

Predation (+, -)

Transfers energy across trophic levels; keeps prey populations under control.

Examples

- **Biological Control:** Moth used to control invasive prickly pear cactus in Australia.
- **Maintaining Species Diversity:** *Pisaster* starfish (predator) removal led to the extinction of >10 invertebrate species due to interspecific competition.

Prey Defenses

- **Camouflage** (insects, frogs), **poisonous** (Monarch butterfly), thorns (*Acacia*, *Cactus*), chemical production (*Calotropis*).

Competition (-, -)

A process in which the fitness of one species (measured by 'r') is significantly lower in the presence of another.

Gause's 'Competitive Exclusion Principle'

Two closely related species competing for the same limiting resources cannot co-exist indefinitely; the competitively inferior one will be eliminated.

Example of Exclusion

- *Abingdon tortoise* became extinct due to competition from introduced goats.

Mechanism for Co-existence

- **Resource Partitioning.** Example: MacArthur's study on five species of warblers on the same tree co-existing due to different foraging patterns.

Population Interactions: Parasitism (+, -), Commensalism (+, 0), Mutualism (+, +)



Parasitism (+, -)

Adaptations: Loss of unnecessary sense organs, suckers/adhesive organs, loss of digestive system, high reproductive capacity.

Types: Ectoparasites (lice on humans) and Endoparasites (human liver fluke).

Special Case: Brood Parasitism. Cuckoo (koel) lays eggs in the nest of the crow. Eggs have evolved to resemble the host's eggs.



Commensalism (+, 0)

Definition: One species benefits, the other is neither harmed nor benefited.

Examples: Orchid as an epiphyte on a mango tree; Barnacles on the back of a whale; Cattle egret and grazing cattle; Clown fish and sea anemone.



Mutualism (+, +)

Definition: Both interacting species benefit.

Examples: Lichens (fungi + algae); Mycorrhizae (fungi + roots of higher plants); **Plant-Pollinator interactions** (Fig and Wasp, Orchid *Ophrys* and bee - 'sexual deceit').



NCERT Line Emphasis

These lines are frequently used to create MCQs, Assertion-Reason, and Statement-based questions.

⚠ NCERT Line

Population ecology is, therefore, an important area because it links ecology to population genetics and evolution.

⚠ NCERT Line

Gause's 'Competitive Exclusion Principle' states that two closely related species competing for the same resources cannot co-exist indefinitely and the competitively inferior one will be eliminated eventually.

⚠ NCERT Line

The female mosquito is not considered a parasite, although it needs our blood for reproduction. (Explanation: It does not live on/in the host for its entire life cycle).

★ Direct NEET Pickup

In nature, a given habitat has enough resources to support a maximum possible number, beyond which no further growth is possible. Let us call this limit as nature's carrying capacity (K)...

★ Direct NEET Pickup

The Mediterranean orchid *Ophrys* employs 'sexual deceit' to get pollination done by a species of bee.

NEET-UG Focus



Previous Year Question Themes

- Identifying growth status from Age Pyramids.
- Calculations based on population growth equation ($N_{t+1} = N_t \dots$).
- Differentiating between J-shaped and S-shaped growth curves and their equations.
- Matching types of population interactions with their signs (+, -, 0) and examples.
- Questions on 'Resource Partitioning' and 'Competitive Exclusion'.



Common Traps & Misconceptions

- **Trap:** Confusing relative density with absolute density.
- **Misconception:** Believing competition only occurs when resources are limited. (NCERT clarifies interference competition can occur even with abundant resources).
- **Trap:** Thinking the female mosquito is a parasite. (It's a predator/sanguivore, using the host for a resource, not as a habitat).



Frequently Confused Terms

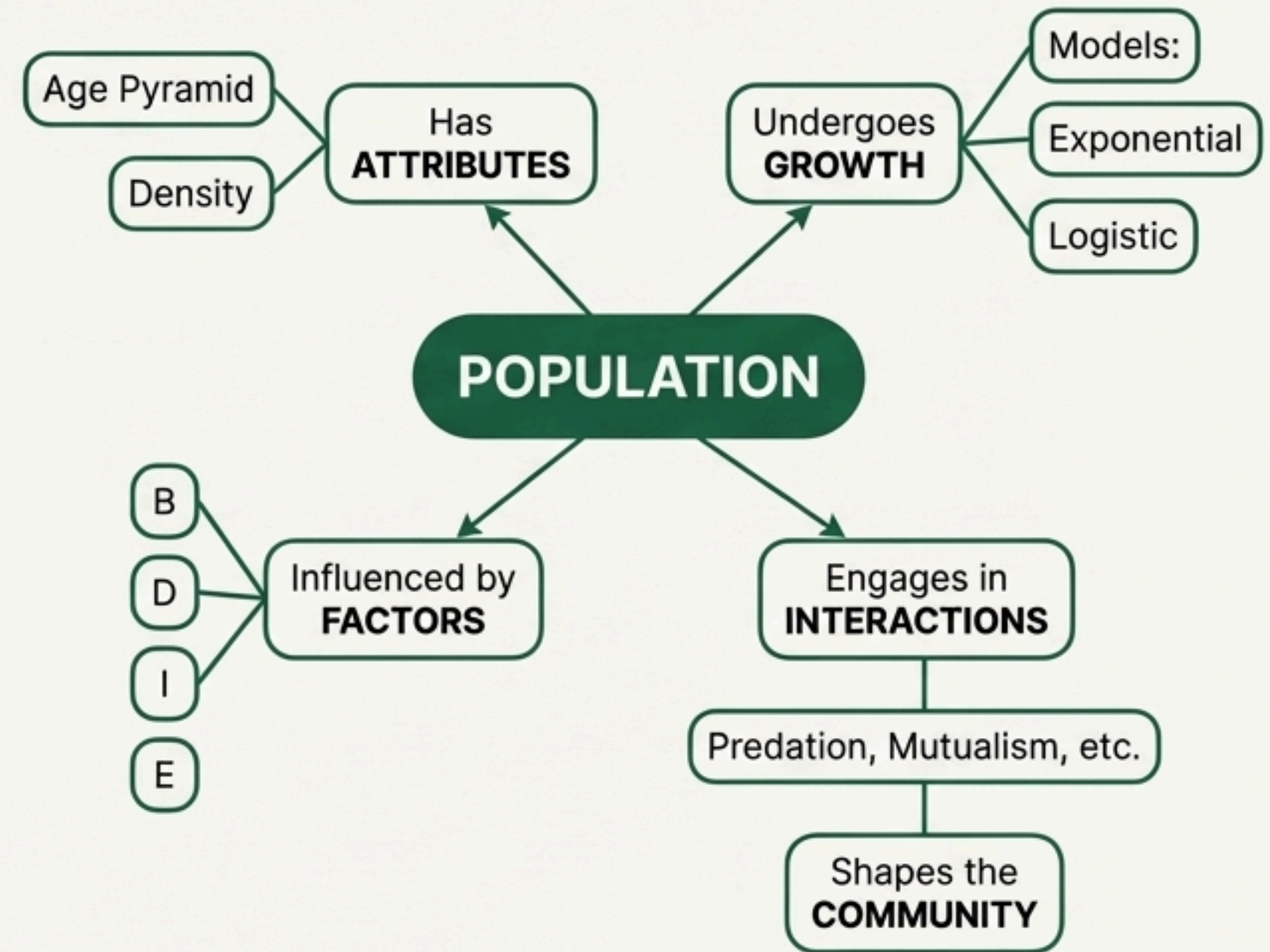
- **Mutualism (+, +)** vs. **Commensalism (+, 0):** Differentiate based on whether the second species benefits or is neutral.
- **Parasitism (+, -)** vs. **Predation (+, -):** Parasites generally live on/in their host and don't kill it immediately, while predators do.

Rapid Revision & Final Chapter Snapshot

Rapid Revision

- **Attributes:** Birth Rate, Death Rate, Sex Ratio, Age Pyramid, Population Density (**N**).
- **Growth Factors:** Natality (**B**), Mortality (**D**), Immigration (**I**), Emigration (**E**).
- **Exponential Growth:** $dN/dt = rN$, J-curve, unlimited resources.
- **Logistic Growth:** $dN/dt = rN((K-N)/K)$, S-curve, limited resources, Carrying Capacity (**K**).
- **Interactions:**
 - Mutualism (+,+)
 - Competition (-,-)
 - Predation (+,-)
 - Parasitism (+,-)
 - Commensalism (+,0)
 - Amensalism (-,0)

Final Chapter Snapshot



If you remember only this much, you can solve most questions.